Md Akib Haider

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Research Interests: Human-Centric AI, Computer Vision, Machine Learning

EDUCATION

- **Islamic University of Technology (IUT)** Aug 2022 – Oct 2026
  B.Sc. in Computer Science and Engineering
  – **CGPA**: 3.91/4.00
- **Notre Dame College (NDC)** 2019–21
  Higher Secondary School Certificate (HSC), Science
  – **GPA**: 5.0/5.0 **(Government Merit Scholarship)**.
- **Ideal School and College (ISC)** 2009–19
  Secondary School Certificate (SSC), Science
  – **GPA**: 5.0/5.0 **(Government Merit Scholarship)**.

RESEARCH EXPERIENCE

- **Prototype-Based Test-Time Prediction Calibration for Few-Shot Object Detection**
  Accepted in WACV | Paper 2027
  – We proposed a Patch-Token Prototype Calibration method by encoding query images via a frozen foundation-model encoder for fully inference-based confidence recalibration of any CNN-based detector.
- **Diagnosing Prototype Estimation in Frozen-Feature Few-Shot Class-Incremental Learning: Training-Free Geometry Selection and Unlabeled-Query Prototype Rectification**
  Accepted in WACV | Paper 2027
  – We proposed finite-shot prototype estimation as the dominant recoverable error source in frozen few shot class-incremental learning, and complements it with a transductive correction for unlabeled query settings.
- **Probabilistic Feature Imputation and Uncertainty-Aware Multimodal Federated Aggregation**
  Published in PMLR | Paper 2026
  – We designed a probabilistic federated learning framework that replaces point estimates with calibrated distributions for uncertainty-aware local fusion and weighted aggregation across missing modalities.
- **LoRA-MME: Multi-Model Ensemble of LoRA-Tuned Encoders for Code Comment Classification**
  Accepted in ICSE workshop | Paper 2026
  – We proposed a weighted stacking ensemble learning method while fine-tuning four complementary code encoders to solve multi-label classification across different programming languages in total 18 classes; achieving better classification performance over baseline.
- **Semi-Supervised Frequency-Aware GAN for Ultra Low to High Field MRI Translation** | Under Review, ISBI
  – We propose a reward-model framework, GRPO with proper-scoring-rule rewards to state both answer confidence and evidence sufficiency, such that model learns to tell an uncertain answer apart from an image that lacks the answer, aiming to reduce confident hallucinations on incomplete visual inputs.
- **Rewarding Calibrated Confidence and Evidence Sufficiency in VLMs** | Work in progress
  – We presented a semi-supervised frequency-aware attention-guided framework for synthesizing 3T MRI from 64mT ultra-low-field scans, addressing paired data scarcity through adversarial refinement.
- **DPPC: Discriminative Pairwise Predictive Correction with Comparative Prompts for Training-Free Few-Shot Class-Incremental Learning** Work in Progress
  – We introduced Tri-Level Calibration (TRIC) in FSCIL, semantic calibration within the language branch, fuses CLIP visual evidence on a common score axis, and inter-modal calibration between resulting visual and semantic evidence for fully inference based classification.

WORK EXPERIENCE

- **Undergraduate Research Intern** Jun 2026 – Present
  University of Illinois Urbana Champaign, SRSE program, Sharifa Sultana's Lab
- **App Design and Development (Remote)** Feb 2025 – Apr 2025
  SwadeshBazar, Chelmsford, UK
  - Built Android and web version in Flutter and Next.js for a grocery delivery and warehousing start-up featuring product browsing, cart, dashboard, designed with Rave and payment gateway for secure transactions.
- **Industrial Training on Mobile Application Development** Sep 2025 – Oct 2025
  Battery Low Interactive Ltd, Dhaka, Bangladesh
  - Built an Anime streaming and sharing app using Flutter and Dart, featuring state management, API integration, UI/UX design, testing, and deployment | **code**.

PROJECTS

- **29 Game State Detection and Move Analysis using Video-Scene Understanding** 2026
  Python, PyTorch, OpenCV, OpenVINO, YOLOv26 | code
  - Built a system to detect game states, analyze gameplay, and predict moves in the card game 29 under low-resource settings.
- **Better_SIS, University Management System** | *Dart, C++, Swift, Flutter* | code 2025
  - Developed Student Information System (SIS) of IUT, featuring academic activities and internal service management for students, teachers, and staff.
- **Student Driven Market Place** | *Flutter, Gemini 2.5, Open Street Map* | code 2025
  - Created a virtual institutional marketplace where students can buy and sell their products within the community, donate items to those in need, and easily manage product deliveries.
- **Animeverse** | *Flutter, Lmstudio, Dart, Docker, Jikan Api, Rive* | code 2025
  - Built an anime streaming platform featuring AI agents, personalized recommendations, anime tracking, discussion forums, downloads, local sharing, and social networking, creating a virtual anime club.
- **Xchange, File Sharing System** | *Socket Programming, Java* | code 2024
  - Developed a secure and efficient peer-to-peer file-sharing system enabling fast file transfers over a local network without requiring internet access.
- **E-Commerce** | *Dart, C++, Rive, Flutter, Docker* | code 2024

GRANTS AND FELLOWSHIPS

- **ACM SIGHPC Travel Grant**
  - HPC Immersion Program, Attended SC 2025 in St. Louis, Missouri, USA ($3000). 2025
  - Lightning Talk on "Data, Model, and Tensor Parallelism for efficient MRI Translation".

ACHIEVEMENTS

Contests & Awards

- **NLBSE Tool Competition** winner, co-located workshop with ICSE-26. 2026
- **Programming Contest**
  - **26th** (among 110 teams), IUT 12th National ICT Fest Programming Contest. 2026
  - **80th** (among 162 teams), Uttara University Inter-University Programming Contest. 2025
  - **6th**, Intra-IUT Junior Programming Contest. 2024
  - **52nd** (among 115 teams), IUT 11th National ICT Fest Programming Contest. 2024
  - **102nd** (among 317 teams), ICPC Asia Dhaka Regional Onsite Contest. 2024
  - Personal best: Country **rank 37**, Global **rank 1536** in Codeforces | profile. 2022-26
- **Hackathon**
  - **Champion** in Coderush 2.0, IUT-ICT Fest Hackathon | code 2025
- **Software Sub-Team Member, Project Altair** at IUT Robotics Society. 2023–24
  - **Global 6th place** and **Best Science Team award** in International Rover Challenge (IRC).

Games & Sports

- **Captain,** Inter University Chess Team of IUT | *FIDE Blitz Rating: 1553*. 2025–26
- All Bangladesh **Champion,** Intra-IUT Annual Drought and Scrabble Competition. 2026
- All Bangladesh **2nd Runner Up,** Intra-IUT Annual Table Tennis Competition. 2026
- All Bangladesh **rank 49** at 37th National Sub-Junior Chess Championship. 2018

Literature & Creative Work

- **Delegate (Writing)** at 6th International Children Film Festival. 2013
- **UNICEF-Meena Media Award** winner for creative writing in print media. 2012–13

LEADERSHIP & EXTRA CURRICULAR ACTIVITIES

- **Instructor, IUT ACM Community** Aug 2024 – Present
  - Mentor for sophomore Competitive Programming aspirants.
  - Organized and conducted weekly classes.
- **Organizer, Intra-IUT Junior Programming Contest** 2025
  - Event head at *Coderush 2.0*, organized by IUT Computer Society.
- **Vice President, Notre Dame English Club** 2019–22
  - Organized *6th National English Carnival*, Literature & Pop-Culture Quiz: *Litbuzz 3.0*.
- **Problem Setting**
  - Intra-IUT Junior Programming Contest | problemset 2025
  - Abid Hasan Intra University Programming Contest | problemset 2025

Sports: Passionate about playing table tennis and chess.

REFERENCES

- **Dr. Md. Hasanul Kabir**
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- **Dr. Sharifa Sultana**
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- **Dr. Rafsanjany Kushol**
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